Introduction to Wireless Communications

Definition and Development of Wireless Communication

Wireless Communications

Wireless Communications

Course Informations

Principles of Wireless Communications —— 无线通信原理

- 面向电子信息类专业学生开设的一门专业课程；
- 了解和掌握无线通信的基本原理、概念、方法及关键技术；
- 初步掌握无线通信系统的性能评价和设计方法；
- 熟悉无线通信专业知识在英语背景中的表达和运用。

Contents

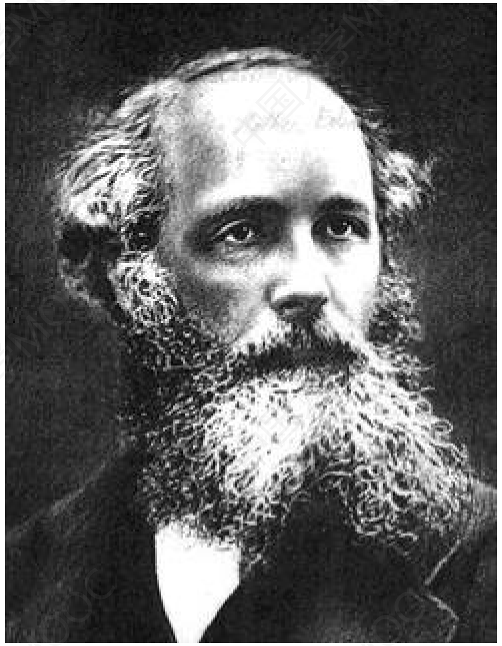
Introduction to Wireless Communications

- **Definition and Development of Wireless Communications**
- Current Wireless Communication Systems
- Mobile communication networks
- Basics of communication and capacity

Definition of Wireless Communications

- **Wireless communication** : The transfer of information over a distance without the use of electrical conductors or "wires".
- **Electromagnetic waves** (usually radio waves) are used in wireless communication to carry the signals.

Devlopment of Wireless Communications



James Clerk Maxwell，
1831.06.13 - 1879.11.05

Heinrich Rudolf Hertz ,
1857.02.22 - 1894.01.01

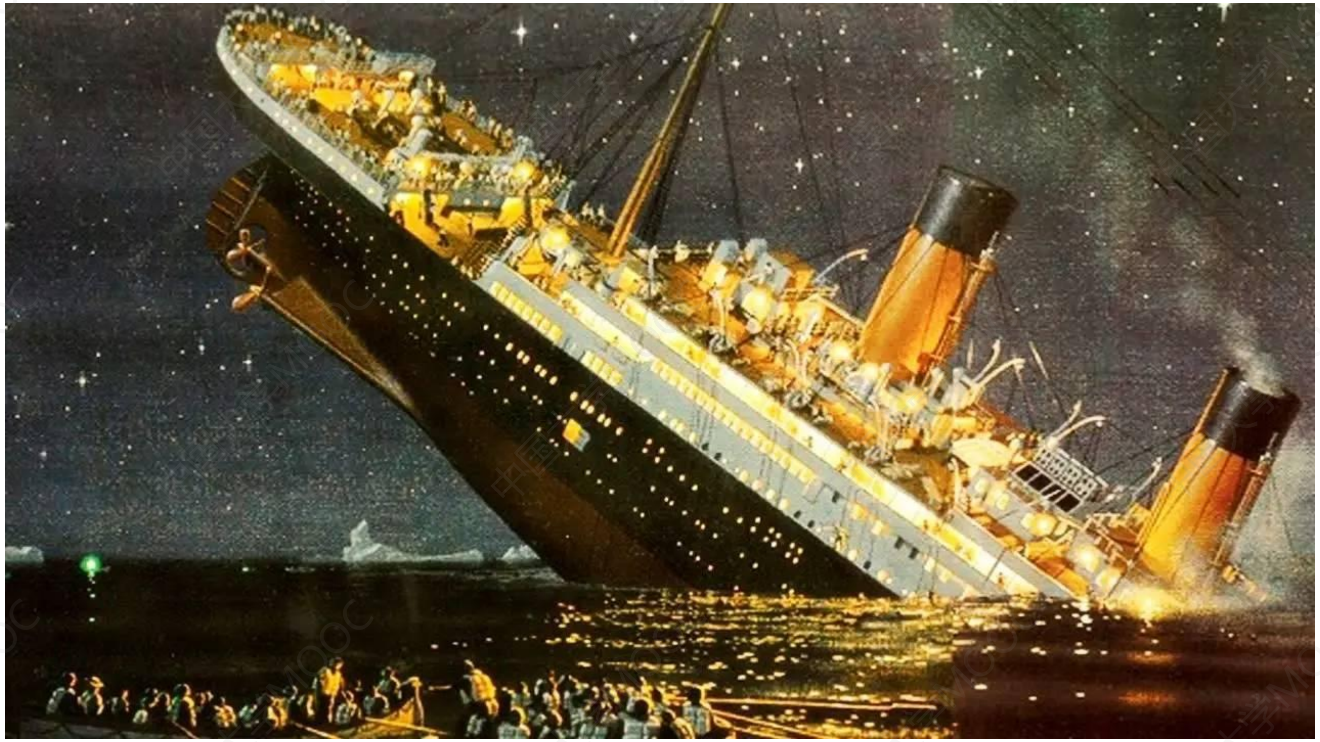
Devlopment of Wireless Communications

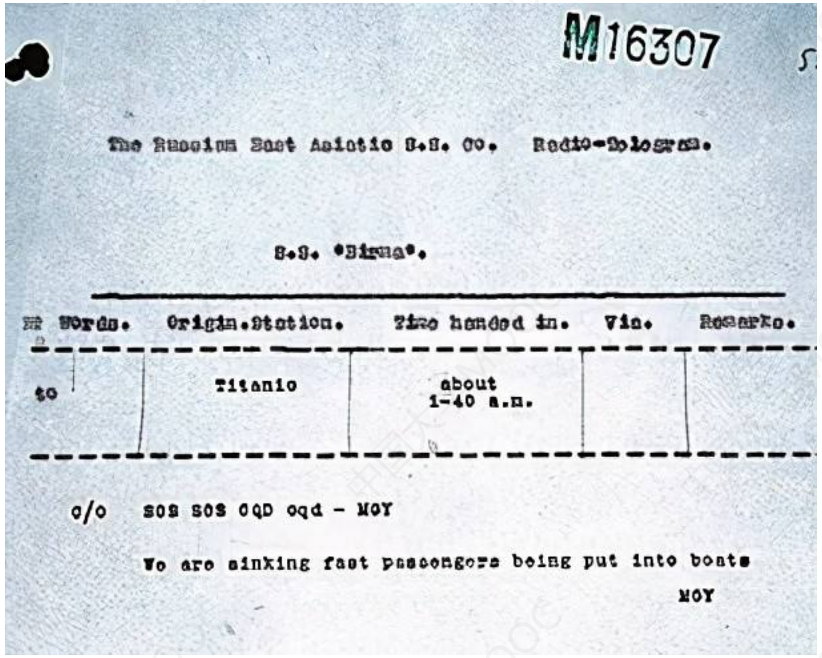
Guglielmo Marconi，1874.04.25 - 1937.07.20.

- 1894-96 First Transmitter – First Patent
- 1897 World’s First Wireless Telegraph & Signal Company
- 1901 Telegraph across the atlantic ocean
- 1909 Nobel prize
- 1920 Discovery of short wave

学习科学家们勇于探索、不断超越自我的精神！

Titanic




M16307

The Russian East Asiatic S.S. Co. Radio-Telegram.

S.S. "Birma".

Words.	Origin.Station.	Time handed in.	Via.	Remarks.
to	Titanic	about 1-40 a.m.		

c/o SOS SOS CQD CQD - MGY

We are sinking fast passengers being put into boats

MGY

Distress telegraph

"Those who had been saved, had been saved through one man, Mr. Marconi... and his marvellous invention."

Band Division of Electromagnetic Waves

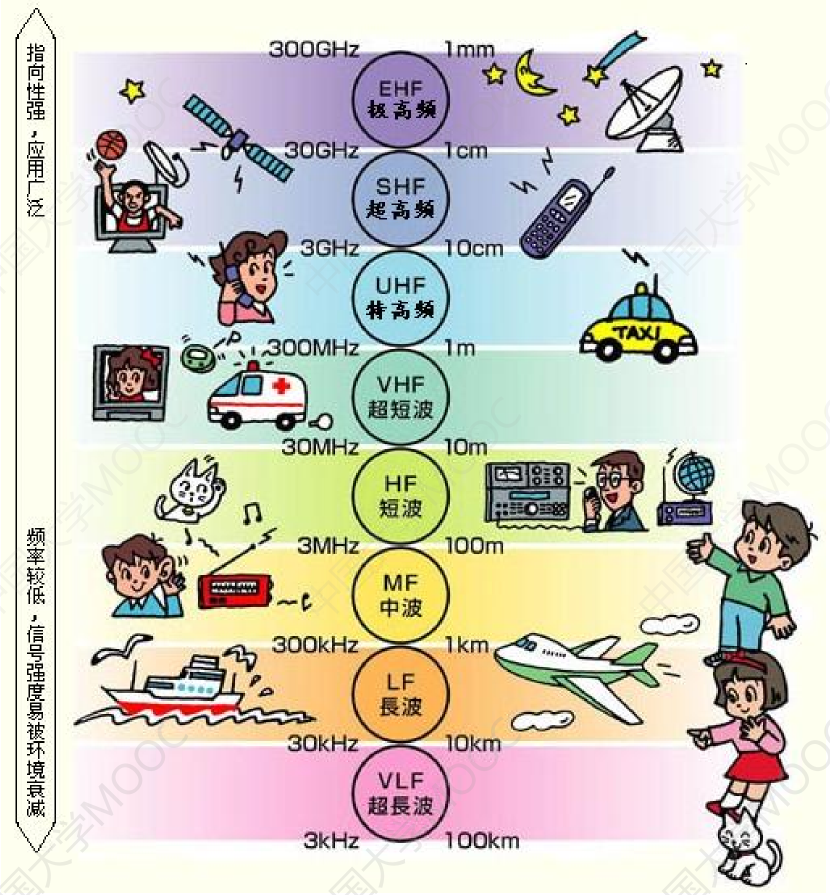
“ frequency ”

$$f = \frac{c}{\lambda}$$

“velocity of light ”

“ wavelength ”

频带名称	频率范围	波段名称	波长范围
至低频（TLF）	0.03-0.3Hz	至长波或千兆米波	10000-1000 兆米（Mm）
至低频（TLF）	0.3-3Hz	至长波或百兆米波	100-100 兆米（Mm）
极低频(ELF)	3-30Hz	极长波	100-10 兆米（Mm）
超低频(SLF)	30-300Hz	超长波	10-1 兆米（Mm）
特低频(ULF)	300-3000Hz	特长波	1000-100 千米(Km)
甚低频(VLF)	3-30KHz	甚长波	100-10 千米(Km)
低频(LF)	30-300KHz	长波	10-1 千米(Km)
中频(MF)	300-3000KHz	中波	1000-100 米(m)
高频(HF)	3-30MHz	短波	100-10 米(m)
甚高频(VHF)	30-300MHz	米波	10-1 米(m)
特高频(UHF)	300-3000MHz	分米波	10-1 分米(dm)
超高频(SHF)	3-30GHz	厘米波	10-1 厘米(cm)
极高频(EHF)(	30-300GHz	毫米波	10-1 毫米(mm)
至高频(THF)	300-3000GHz	丝米波或亚毫米波	10-1 丝米(dmm)

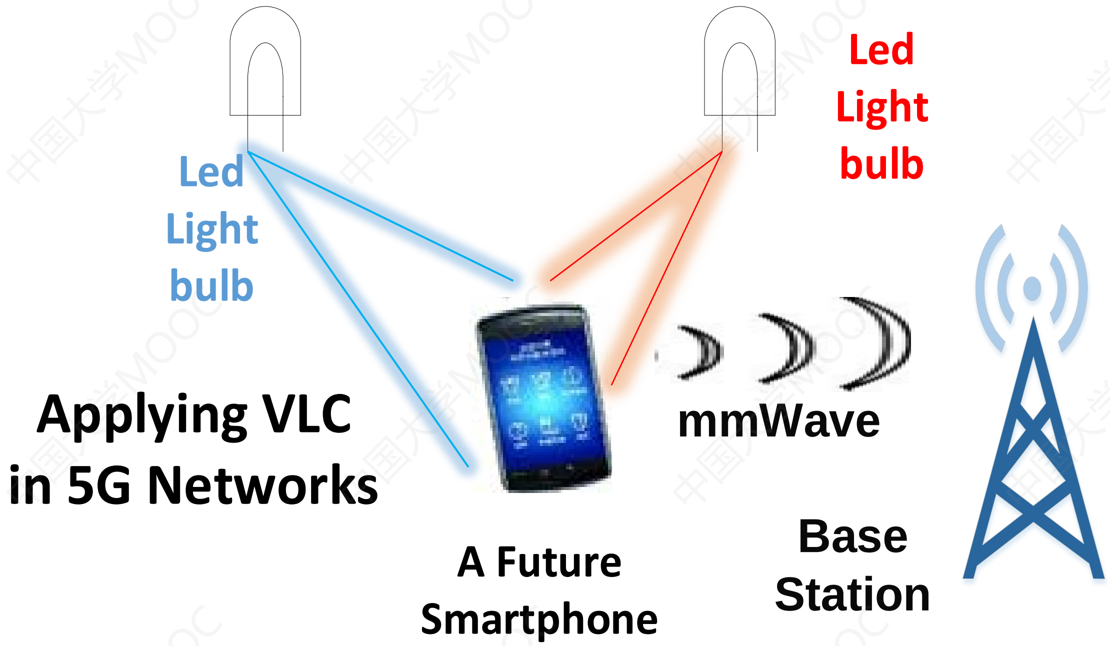


How do cell phones work ?

“ Radio Frequency (RF) ”

- **Radio wave, one type of electromagnetic waves, is used for such form of wireless communication.**
- **Most cell phones operate in the frequency range of 800 to 2,200 Megahertz.**

What frequency do cell phones operate on?



The mmWave and visible light communications (VLC) are two promising technologies for future 5G indoor communications.

What frequency do cell phones operate on?

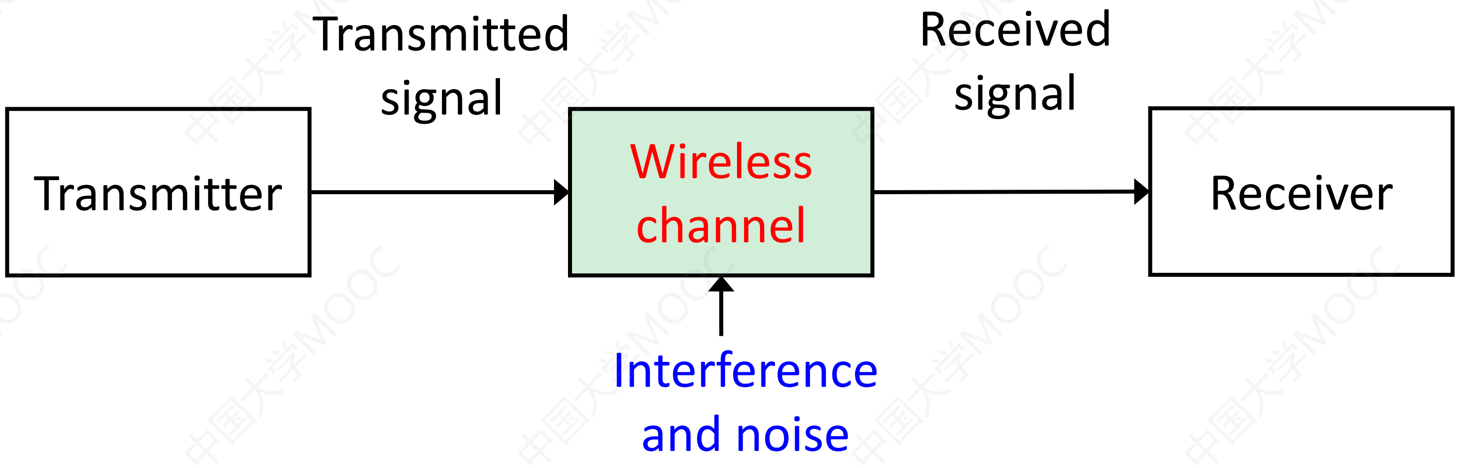
- **FR1** : Sub-6 GHz (4.5 GHz – 6 GHz)
- **FR2** : mmWave (24.2 GHz – 52.6 GHz)

For example, if a carrier frequency *f* = 30 GHz is adopted for the 5G New Radio, the wavelength *λ* of the carrier wave is

$$\lambda = \frac{C}{f} = \frac{3\times 10^8}{30\times 10^9} = 10\ (\mathrm{mm})$$

- The higher the frequency, the higher the potential speed of the signal.
- The higher the frequency, the larger the propagation loss.

Wireless Vs. Wire Communications



- **The channel is a medium through which the transmitter output is sent.**
- **Based on the channel type, communication systems can be divided into wire communication systems and wireless communication systems.**

Wireless Vs. Wire Communications

Challenges in Wireless Communication Systems

- **Information is transmitted by radio waves**

  Radio channel is not a good transmission channel:

  - **Large path loss**
  - **Multipath**
  - **Time-varing**

- **In complex interference environment**

  Interferences from

  - **Space (cosmic, atmosphere phenomenon)**
  - **Industry equipment**
  - **Other electronic and communication systems**
  - **Themselves such as multiuser interference (MUI).**

Although wireless is much less reliable than wire, it is inevitable !

Thanks !